

6. A POSET is called a _____ lattice if all its subsets have both a join and a meet.
7. Hasse diagram are drawn for POSET. (TRUE/FALSE)
8. Linear trend of a time series indicates towards constant rate of change. (TRUE/FALSE)
9. The component of a time series attached to long-term variations is termed as _____.
10. A time series data is a set of data recorded at Periodically. (TRUE/FALSE)
11. If $f(x) = y$ then $f^{-1}(y)$ is equal to _____
12. A function is said to be _____ if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .
13. Time horizon is not a forecasting technique. (TRUE/FALSE)
14. A _____ graph is a graph that can be embedded in the plane.
15. A graph is a collection of _____ and _____.
16. The origin and terminal of a walk are same is called _____.

Q-2 Answer the following questions (Any Seven)

[14]

1. Define Set, Finite and Infinite Set.
2. Explain the rules of sum and product.
3. Define Lattice. List the basic properties of Lattices.
4. Explain Complete Lattice and Bounded Lattice.
5. Define Graph and Multi Graph with suitable examples.
6. Explain Tractable and Intractable Problems.
7. Define Time Series with utilities provided by Time Series analysis.
8. List and Explain Components of Time Series.
9. Define Secular Trend. List the uses of secular trend.

Q-3 Answer the following Questions:

- A. Define binary relation. Also explain reflexive, symmetric and anti-symmetric relations with suitable examples.

[08]

OR

- B. [i] Let $A = \{1, 2, 3\}$. Then investigate the following relations for reflexive, symmetric, antisymmetric, or transitive.

[08]

$$[1] R = \{(1,1), (2,2), (3,3), (1,3), (1,2)\}$$

$$[2] R = \{(1,1), (2,2), (1,3), (3,1)\}$$

$$[3] R = \{(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2)\}$$

[ii] Define POSETS. Draw the Hasse diagram of $\langle P(A), \subseteq \rangle$ for $A = \{a, b, c\}$ and $B = \{a, b, c, d\}$

Q-4 Answer the following Questions:

[08]

- A. Discuss Lattices and Explain meet and Join operations.

OR

- B. Explain Euler and Hamilton Path, Circuit with suitable example.

[08]

Q-5 Answer the following Questions: [08]

A. Define Algorithm to find largest number among given n numbers with its time complexity.

OR

B. Explain Dijkstra (shortest path) algorithm in detail. [08]

Q-6 Answer the following Questions: [08]

A. Below is given the COVID 19 pandemic cases in INDIA (JAN – AUG)

Month	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG
Cases (in thousands)	80	90	92	83	94	99	92	104

Fit a straight line trend equation by the method of least squares and estimate the trend values.

OR

B. Fit a straight line trend by the method of Least Squares. Calculate the Trend values. Estimate the value for 2022. [08]

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
Earnings (Rs. Lakh)	4	7	7	8	9	11	13	14	17
